

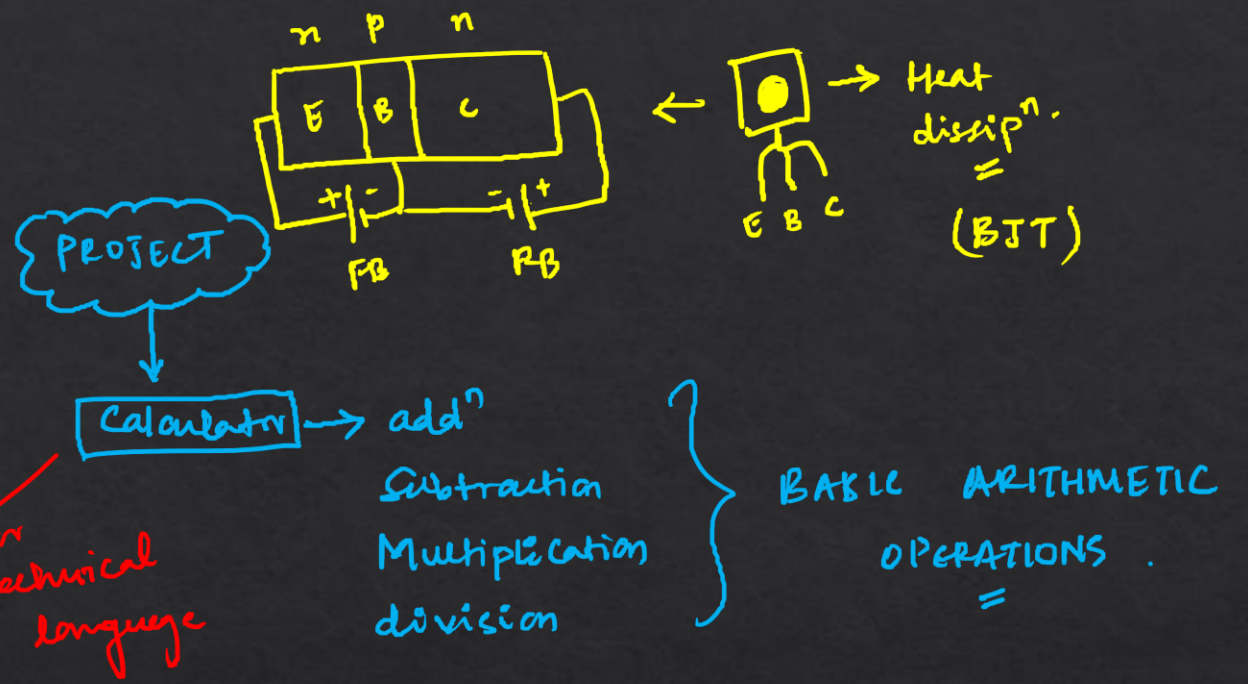
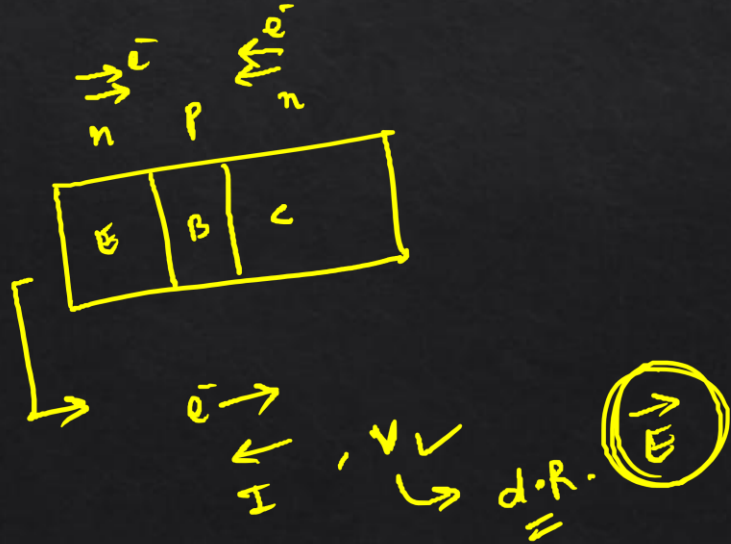
VLSI CLASSES Silicon [17-11-2024]

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1. HDL (Hardware description language).
2. Semiconductor physics ✓
3. DIODE → TRIODE → TETRODE
4. RC chas

$\Rightarrow \text{MOBILITY} \propto \frac{1}{m_p}$
 $\Rightarrow \mu_e \gg \mu_h$
 $\Rightarrow m_e \ll m_p$
 $\Rightarrow \boxed{\mu_e \gg \mu_h}$

$m_e = 9.1 \times 10^{-31} \text{ kg}$
 $m_p = 1.67 \times 10^{-27} \text{ kg}$

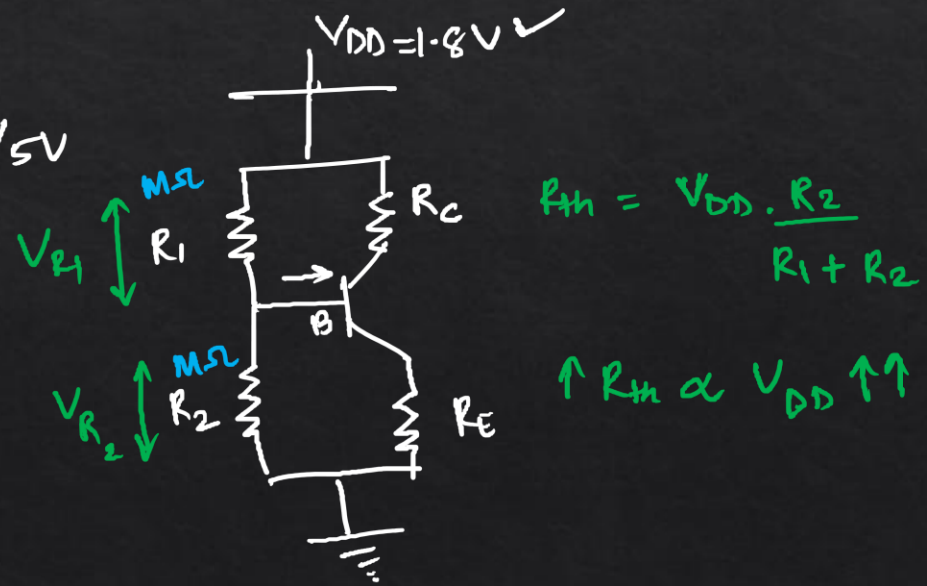


DIGITAL SYSTEM DESIGN

Specifications → $V_{DD} = 3.3V / 1.8V / 5V$

Mobility
 $\mu_e = \frac{V_d}{E}$

$\phi = CV$



DIGITAL SYSTEM DESIGN.

[CALCULATOR]

C lang. ✓
statich ✓
math.h ✓

variables
↓
int a, b → sum = a + b;
diff = a - b;
div = a ÷ b;
mul = a · b;



$a \cdot b = a \times b = 0 \times 1 = 0$
 $= 1 \times 0 = 0$
 $= 1 \times 1 = 1$

adder = 1+1+1+1 = 5
repeated add = 5 x 1 = 5

OR gate = add → A+B = 1+1 = 1 : (H) x 3 → repeated add
Sub → A-B = 1-1 = 0 : (L)

PROGRAM
= corresponding
ckt

corresponding
digital
ckt

* C lang → Normal programming language

* Verilog / System Verilog / VHDL / FPGA → HDL → SYNTHESIZABLE → HARDWARE
tb design
DESCRIPTION lang.
Simulation → AEC → BJT

CALCULATOR = $\rightarrow$ Technical World [VLSI Company].

SYSTEM SPECIFICATIONS $\rightarrow V_{dd}, gnd, R(k\Omega, M\Omega)$
functionalities
 $+, -, *, \div, \%$

FUNCTIONAL ARCHITECTURE DESIGN

FUNCTIONAL VERIFICATION

LOGIC DESIGN

LOGIC VERIFICATION

CKT DESIGN ✓

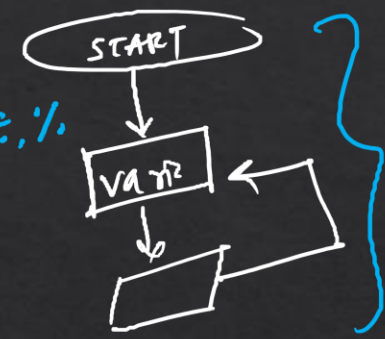
CKT VERIFICATION

LAYOUT DESIGN

LAYOUT VERIFICATION

FABRICATION + TESTING ✓

PACKAGING. ✓



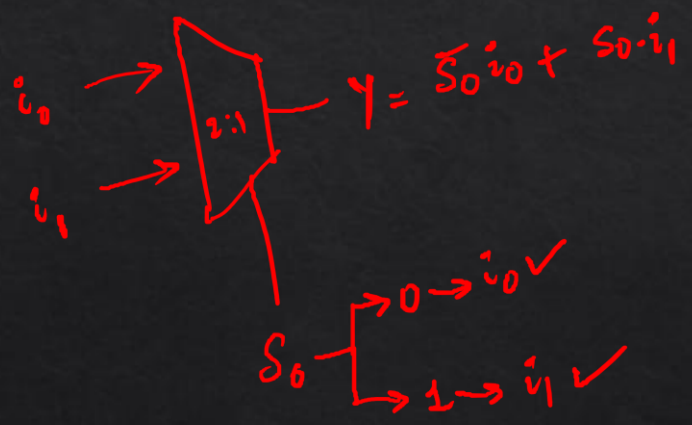
FAD = algorithm
 $\downarrow$
No. of instructions.
=

$1+1=0$ ✗

$sum = a + b$ $(1+1=1)$ $(2+3=5)$ ✓

$sum = a - b = diff$ ✗

MUX = 2:1



$\rightarrow$ PURELY ANALOG JOB (Backend).

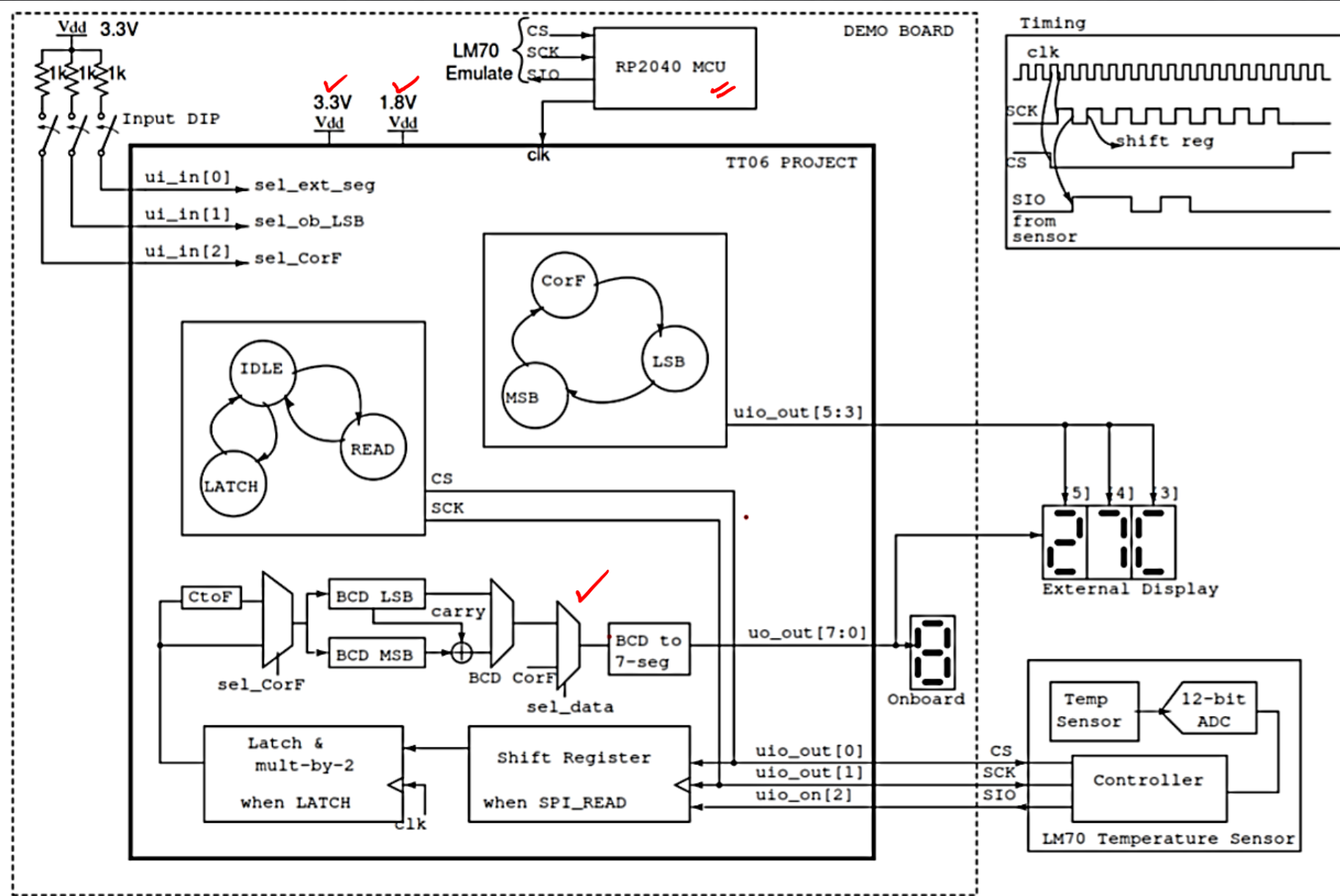


Figure 1: Block diagram of the complete system.

→ FUNCTIONAL ARCH. DESIGN

- ① $\bar{S}_0 i_0 + S_0 i_1$
 $0 \uparrow i_0$
- ② $2 : 1 \rightarrow$ No. of selection lines
 $1 \rightarrow$ No. of i/p
- No. of o/p

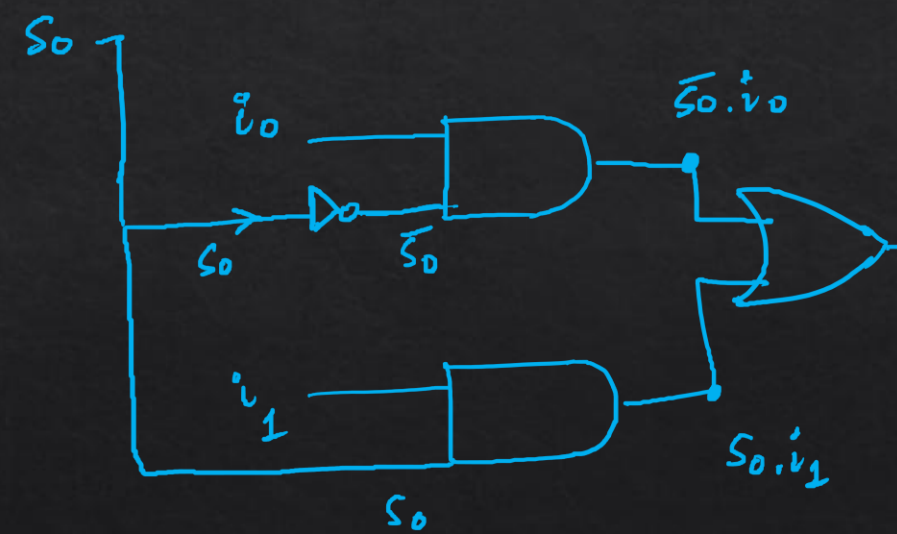
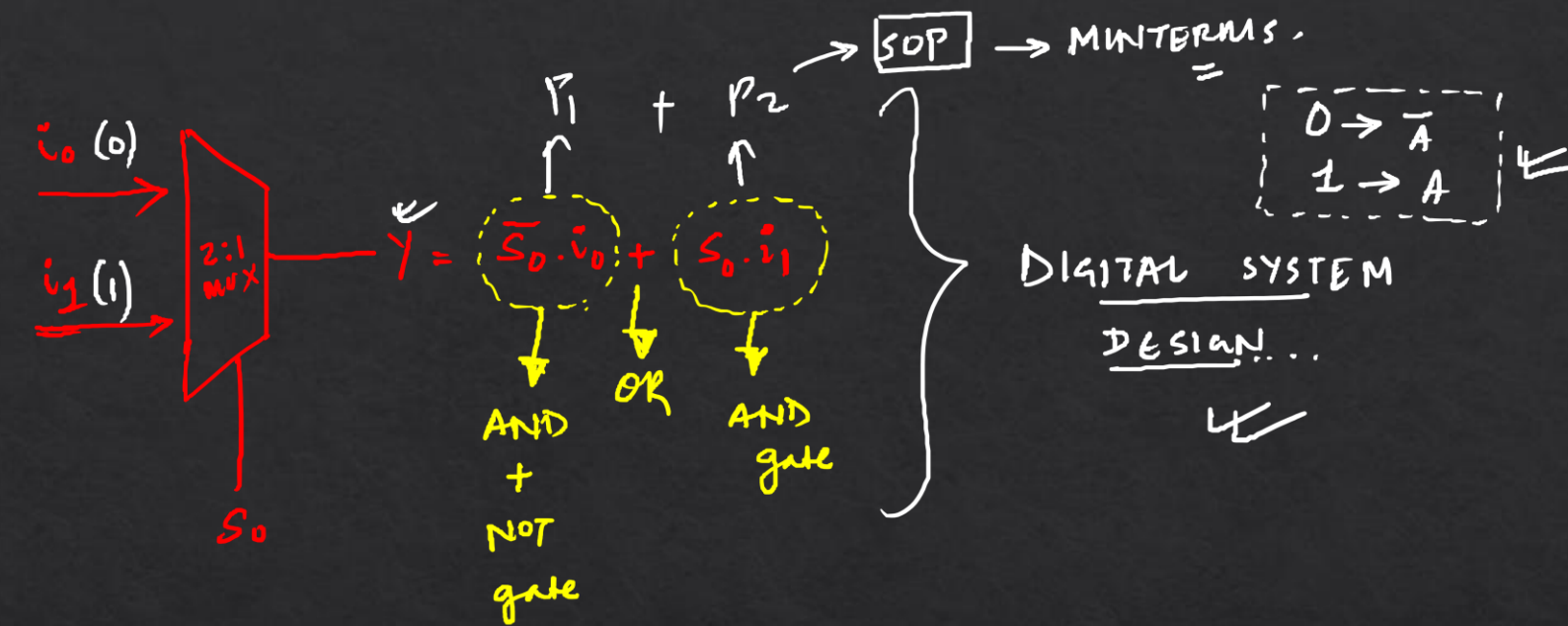
MUX

2:1 MUX

- ① $2^1 : 1$
 $\rightarrow 0, 1$

- ② $n : 1$ MUX $\rightarrow n = ?$

$2^2 : 1 \rightarrow 4 \rightarrow 0 \rightarrow 2^n - 1 \rightarrow 00, 01, 10, 11$



2^3	2^2	2^1	2^0		
S_1	S_0	Y			dec
0	0	-	$i_0 \rightarrow$	(0)	
0	1	-	$i_1 \rightarrow$	(1)	
1	0	-	$i_2 \rightarrow$	(2)	
1	1	-	$i_3 \rightarrow$	(3)	

MSB $\leftarrow$ $\rightarrow$ LSB

$$\overline{A+B} = \overline{A} \cdot \overline{B} = \overline{A \cdot B}$$

2:1

4:1 MUX

(0 → 3)

16:1 MUX

2:1 MUX

(0 → 15)

z^3	z^2	z^1	z^0		
s_1	s_0	y			dec.
0	0	i_0	→	(0)	
0	1	i_1	→	(1)	
1	0	i_2	→	(2)	
1	1	i_3	→	(3)	

MSB ← → LSB

s_3	s_2	s_1	s_0	y	dec.
0	0	0	0	i_0	0
⋮	⋮	⋮	⋮	⋮	⋮
1	1	1	1	i_{15}	15

SOP → POS → SOP

de-morgan's law de-morgan's law

00, 01, 10, 11

$$Y = \overline{s_1} \overline{s_0} i_0 + \overline{s_1} s_0 i_1 + s_1 \overline{s_0} i_2 + s_1 s_0 i_3$$

$$\downarrow$$

$$(\overline{s_1} + \overline{s_0} + i_0) \cdot (\overline{s_1} + \overline{s_0} + i_1) \cdot (s_1 + \overline{s_0} + i_2) \cdot (s_1 + s_0 + i_3)$$

$$s_3 s_2 s_1 s_0 i_{15}$$